

1번

```
clear; clc;

% (a)
t = 0:0.01:30;
f = exp(-0.12*t).*sin(5*t) + 0.3*cos(2*t);

figure
plot(t, f)
grid on
xlabel('t')
ylabel('f(t)')
title('1번 신호 그래프')

% (b)
u = exp(-0.12*t) + 0.3;
l = -exp(-0.12*t) - 0.3;

hold on
plot(t, u)
plot(t, l)

% (c)
idx = find(f(2:end-1) > f(1:end-2) & f(2:end-1) > f(3:end)) + 1;

% (d)
plot(t(idx), f(idx), 'o')
legend('f(t)', 'u(t)', 'l(t)', 'local max')

% (e)
local_max_result = [t(idx)' f(idx)']
```

2번

```
clear; clc;

% (a)
x = -5:0.05:5;
y = -5:0.05:5;

[X, Y] = meshgrid(x, y);
Z = sin(X.*Y) ./ (1 + X.^2 + Y.^2);

% (b)
figure
mesh(X, Y, Z)
grid on
xlabel('x')
ylabel('y')
```

```

xlabel('z')
title('2번 3차원 그래프')

% (c)
figure
contour(X, Y, Z, 25)
grid on
xlabel('x')
ylabel('y')
title('2번 등고선 그래프')

% (d)
absZ = abs(Z);
max_abs_Z = max(absZ(:))

% (e)
idx = find(absZ == max_abs_Z);
[row, col] = ind2sub(size(absZ), idx(1));

max_x = X(row, col)
max_y = Y(row, col)

```

3번

```

clear; clc;

% (a)
A = [7 2 1;
     2 6 -2;
     1 -2 5];

[V, D] = eig(A)

% (b)
eigenvalues = diag(D)

% (c)
[lambda_max, index] = max(eigenvalues)

% (d)
v = V(:, index)

% (e)
unit_v = v / norm(v)

% (f)
check = A*v - lambda_max*v

```

4번

```

clear; clc;

X = [2 5 1;
     4 8 3;
     6 11 7;
     9 15 10;
     13 20 14];

% (a)
mean_X = mean(X)

% (b)
Xc = X - mean_X

% (c)
S = Xc' * Xc

% (d)
[V, D] = eig(S);
eigenvalues = diag(D)

% (e)
[lambda_max, index] = max(eigenvalues)

% (f)
main_vector = V(:, index)

```

5번

```

clear; clc;

% (a)
A = [5 2 -1;
     -2 6 3;
     1 -1 7];

b = [8;
     5;
     12];

% (b)
x1 = inv(A) * b

% (c)
x2 = A \ b

% (d)
difference = x1 - x2

% (e)

```

```

det_A = det(A)

if det_A ~= 0
    result = 'det(A)가 0이 아니므로 해가 유일하게 존재한다'
else
    result = 'det(A)가 0이므로 해가 유일하게 존재하지 않는다'
end

```

6번

```

clear; clc;

% (a)
x = -6:0.05:6;
y = -6:0.05:6;

[X, Y] = meshgrid(x, y);

Z = exp(-0.08*(X.^2 + Y.^2)) .* cos(3*X) .* sin(2*Y);

% (b)
figure
surf(X, Y, Z)
grid on
xlabel('x')
ylabel('y')
zlabel('z')
title('6번 3차원 표면 그래프')

% (c)
figure
contour(X, Y, Z, 30)
grid on
xlabel('x')
ylabel('y')
title('6번 등고선 그래프')

% (d)
max_Z = max(Z(:))
min_Z = min(Z(:))

% (e)
max_idx = find(Z == max_Z);
[max_row, max_col] = ind2sub(size(Z), max_idx(1));

min_idx = find(Z == min_Z);
[min_row, min_col] = ind2sub(size(Z), min_idx(1));

max_x = X(max_row, max_col)
max_y = Y(max_row, max_col)

```

```
min_x = X(min_row, min_col)
min_y = Y(min_row, min_col)
```

7번

```
clear; clc;

n = [50, 100, 200, 400, 800, 1600];
T = [14, 19, 30, 51, 92, 170];

% (a), (b), (c), (d)
figure

subplot(1,3,1)
plot(n, T, 'o-')
grid on
xlabel('n')
ylabel('T')
title('일반 그래프')

subplot(1,3,2)
semilogx(n, T, 'o-')
grid on
xlabel('n')
ylabel('T')
title('x축 로그 그래프')

subplot(1,3,3)
semilogy(n, T, 'o-')
grid on
xlabel('n')
ylabel('T')
title('y축 로그 그래프')

% (e)
selected_graph = '일반 그래프(plot)가 가장 직선에 가깝게 보인다'
```

8번

```
clear; clc;

% (a)
t = 0:0.01:24;

P = 50 + 18*sin(pi*t/12) + 6*cos(pi*t/4);

figure
plot(t, P)
```

```

grid on
xlabel('t')
ylabel('P(t)')
title('8번 하루 전력 사용량')

% (b)
[max_P, max_index] = max(P)

% (c)
max_time = t(max_index)

% (d)
[min_P, min_index] = min(P)
min_time = t(min_index)

% (e)
hold on
plot(max_time, max_P, 'o')
plot(min_time, min_P, 's')
legend('P(t)', 'maximum', 'minimum')

```

9번

```

clear; clc;

X = [1 3 0;
     2 5 1;
     4 8 3;
     7 12 5;
     11 17 9];

% (a)
x1 = X(:,1);
x2 = X(:,2);
x3 = X(:,3);

% (b)
theta12 = acosd(dot(x1, x2) / (norm(x1) * norm(x2)))

% (c)
theta13 = acosd(dot(x1, x3) / (norm(x1) * norm(x3)))
theta23 = acosd(dot(x2, x3) / (norm(x2) * norm(x3)))

% (d)
theta = [theta12 theta13 theta23]

% (e)
[theta_min, theta_index] = min(theta)

pairs = [1 2;

```

```
1 3;  
2 3];
```

```
sensor_pair = pairs(theta_index,:)
```

10번

```
clear; clc;  
  
% (a)  
A = [9 -2 -1;  
     -2 8 -3;  
     -1 -3 10];  
  
b = [15;  
     7;  
     11];  
  
% (b)  
I = A \ b  
  
% (c)  
det_A = det(A)  
  
if det_A ~= 0  
    result = 'A는 가역행렬이다'  
else  
    result = 'A는 가역행렬이 아니다'  
end  
  
% (d)  
abs_I = abs(I)  
  
% (e)  
[max_current, current_index] = max(abs_I)
```

11번

```
clear; clc;  
  
% (a)  
t = 0:0.01:40;  
  
y = exp(-0.04*t) .* (sin(4*t) + 0.6*sin(9*t));  
  
figure  
plot(t, y)  
grid on  
xlabel('t')  
ylabel('y(t)')
```

```

title('11번 출력 신호')

% (b)
u = 1.6 * exp(-0.04*t);
l = -1.6 * exp(-0.04*t);

% (c)
figure
plot(t, y)
hold on
plot(t, u)
plot(t, l)
grid on
xlabel('t')
ylabel('y(t)')
title('11번 y(t), u(t), l(t)')
legend('y(t)', 'u(t)', 'l(t)')

% (d)
figure
semilogy(t, abs(y))
grid on
xlabel('t')
ylabel('|y(t)|')
title('11번 로그 스케일 그래프')

% (e)
u_start = u(1)
u_end = u(end)

```

12번

```

clear; clc;

epoch = [1, 2, 5, 10, 20, 50, 100, 200];
loss = [3.1, 2.4, 1.7, 1.2, 0.83, 0.36, 0.14, 0.06];

% (a)
figure
plot(epoch, loss, 'o-')
grid on
xlabel('epoch')
ylabel('loss')
title('12번 일반 그래프')

% (b)
figure
semilogy(epoch, loss, 'o-')
grid on
xlabel('epoch')

```



```

ylabel('loss')
title('12번 로그 스케일 그래프')

% (c)
r = loss(2:end) ./ loss(1:end-1)

% (d)
mean_r = mean(r)

% (e)
result_text = ["평균 비율이 1보다 작지 않으므로 loss가 전반적으로 감소한다고 보기 어렵다";
               "평균 비율이 1보다 작으므로 loss가 전반적으로 감소한다"];

result = result_text((mean_r < 1) + 1)

```